# NIRIKSHAN: Product Requirements Document

*AI-assisted building deterioration monitoring and inspection prioritisation for Mumbai. Prepared for Seva First Innovation Challenge 2026. Version 0.1, 8 October 2026.*

## 1. Summary

NIRIKSHAN turns periodic smartphone photographs of ageing buildings into a longitudinal record of visible deterioration, and ranks buildings by how urgently a structural engineer should revisit them. It does not declare any building safe or unsafe. It directs limited engineering attention to the buildings showing the most concerning visible change between formal inspections.

**One-line pitch:** MHADA flags about 80 buildings a year after its pre-monsoon survey. NIRIKSHAN tracks visible deterioration between surveys and tells engineers which buildings to revisit first.

## 2. Problem

Mumbai's Building Repairs and Reconstruction Board (MBRRB, under MHADA) runs an annual pre-monsoon survey of old cessed buildings. In 2026 it identified 82 highly dangerous buildings in South Mumbai, housing 2,736 tenants. 43 of those 82 were also on the previous year's list. A 2020 report put the city's cessed buildings at more than 14,000, and noted that the board runs an emergency control room for resident distress calls.

The gap this product addresses:

- Formal inspections are periodic. Conditions can change between them, especially across a monsoon.
- Engineers cannot visit thousands of buildings frequently.
- Residents notice changes but have no structured way to report them, and no way to show what has changed since last time.
- There is no cheap, consistent record of how a building's visible condition has evolved.

*Figures above are from published news reports and should be re-verified against MHADA's own notices before use in the final submission.*

# 3. Goals and non-goals

## Goals

1. Detect and measure visible distress (cracks, spalling, exposed reinforcement, moisture damage) from smartphone images.
2. Detect **change** in that distress between visits to the same building element, separating real deterioration from lighting, camera and weather differences.
3. Produce an **Inspection Priority Score** and a ranked review queue for engineers.
4. Keep a per-building timeline that engineers can audit.
5. Work with zero specialised hardware.

## Non-goals

- Predicting collapse or assigning a structural safety rating.
- Replacing structural audits or engineer judgement.
- Assessing interior structural elements, foundations or hidden corrosion that are not visible in photographs.
- Drone capture or 3D reconstruction in the MVP (roadmap only).

# 4. Users

| Persona | Needs | Primary use |
|---|---|---|
| **Inspection engineer** (MBRRB, MHADA, BMC) | Know which buildings to visit first, see evidence of change | Dashboard, review queue, building timeline |
| **Field capturer** (junior engineer, trained volunteer, NSS/NCC student) | Capture consistent, usable photos quickly | Mobile capture app with guided overlay |
| **Resident or society member** (later phase) | Report a change and be taken seriously | Simple report flow routed to human review |
| **Administrator** | Manage buildings, users, thresholds | Admin console |

# 5. Scope

## MVP (target: Build stage, 12 Oct to 10 Nov 2026)

- Building registration with baseline photo set
- Guided capture with previous-photo overlay
- Crack and spalling segmentation
- Image alignment between visits
- Change metrics for crack extent and spalling area
- Inspection Priority Score and ranked queue
- Engineer dashboard with building timeline
- Evaluation on staged and archival data

## Phase 2

- Exposed reinforcement detection
- Moisture and efflorescence detection
- Balcony, slab edge and chajja-specific analysis
- Monsoon Watch recheck list
- Citizen reporting into a human review queue

## Phase 3

- Structure-from-motion façade reconstruction
- Authorised drone imagery ingestion
- Integration with MBRRB workflows and control room

# 6. Core user flows

## 6.1 Register a building

1. Admin or engineer creates a building record: ID, address, ward, age, previous inspection category (if known), notes.
2. Field capturer takes a baseline set of photos at defined capture points (for example, each façade, selected balconies, selected columns).
3. The app checks image quality and stores each photo with its capture point label.

## 6.2 Periodic capture

1. Capturer selects the building and capture point.

2. The baseline or previous photo appears as a semi-transparent overlay.
3. The app shows a live alignment indicator and blocks capture when quality or alignment is below threshold, with a clear reason (blur, too dark, too far).
4. Photo is uploaded when connectivity returns.

### 6.3 Analysis

1. System aligns the new image to the previous one.
2. Damage models segment cracks and spalling in both.
3. Change module compares aligned regions and outputs per-region change metrics.
4. Priority score is recalculated for the building.

### 6.4 Engineer review

1. Engineer opens the ranked queue.
2. Selects a building and sees before/after overlays with detected changes highlighted.
3. Marks each finding as confirmed, false alarm or needs site visit.
4. Feedback is stored and used for evaluation and retraining.

## 7. Functional requirements

Priority key: **P0** MVP must-have, **P1** MVP if time allows, **P2** later.

### 7.1 Capture app

| ID | Requirement | Priority |
|---|---|---|
| CAP-1 | Capture photos with building ID, capture point, timestamp, and GPS (if permitted) | P0 |
| CAP-2 | Show previous photo as adjustable overlay during capture | P0 |
| CAP-3 | Real-time quality check for blur, exposure and distance | P0 |
| CAP-4 | Allow a reference scale marker in frame (optional) to support approximate sizing | P1 |
| CAP-5 | Offline capture with later sync | P1 |
| CAP-6 | Blur faces and number plates on device before upload | P1 |

## 7.2 Damage detection

| ID | Requirement | Priority |
|---|---|---|
| DET-1 | Segment crack pixels and output length, area, orientation and branching | P0 |
| DET-2 | Segment spalling regions and output area | P0 |
| DET-3 | Segment exposed reinforcement | P2 |
| DET-4 | Segment moisture staining and efflorescence | P2 |
| DET-5 | Report a confidence value per detection | P0 |

Crack width is reported only as a relative class (fine, moderate, wide) unless a scale marker is present. The system must not present phone-derived width as a precise measurement.

## 7.3 Change detection

| ID | Requirement | Priority |
|---|---|---|
| CHG-1 | Register new and previous images of the same capture point using feature matching and homography | P0 |
| CHG-2 | Normalise illumination before comparison | P0 |
| CHG-3 | Output new, extended and unchanged damage regions | P0 |
| CHG-4 | Reject comparisons when alignment confidence is below threshold and request recapture | P0 |
| CHG-5 | Suppress changes explained by shadows, rain streaks, vegetation, and clutter | P1 |
| CHG-6 | Learned change-detection model (for example, Siamese network) as an alternative to rule-based differencing | P2 |

## 7.4 Inspection Priority Score

| ID | Requirement | Priority |
|---|---|---|
| SCR-1 | Compute a 0 to 100 score from observed change features, weighted by configurable weights | P0 |
| SCR-2 | Map score to bands: 0–30 Routine, 31–60 Review recommended, 61–80 High, 81–100 Urgent human review | P0 |
| SCR-3 | Show the top contributing factors for every score | P0 |
| SCR-4 | Incorporate verified metadata such as building age, previous inspection category and repair history | P1 |
| SCR-5 | Account for rate of change over time | P1 |

The score is labelled "Inspection Priority". The product must never use the terms "collapse probability", "safe" or "unsafe".

## 7.5 Dashboard

| ID | Requirement | Priority |
|---|---|---|
| DSH-1 | Ranked queue of buildings with score, band and reason summary | P0 |
| DSH-2 | Building page with timeline of scores and photos | P0 |
| DSH-3 | Side-by-side and overlay before/after views with detected changes highlighted | P0 |
| DSH-4 | Map view coloured by band | P1 |
| DSH-5 | Engineer feedback controls (confirm, false alarm, needs visit) | P0 |
| DSH-6 | Export a building report as PDF | P1 |

## 7.6 Citizen reporting and Monsoon Watch (Phase 2)

| ID | Requirement | Priority |
|---|---|---|
| CIT-1 | Residents can submit photos with a short description | P2 |

| ID | Requirement | Priority |
|---|---|---|
| CIT-2 | Submissions are matched to a building and go to a human review queue, never straight to a score | P2 |
| MON-1 | Before and after heavy-rain periods, generate a recheck list from recent change and rainfall data | P2 |

## 8. AI and ML requirements

- **Models:** start with a segmentation model for cracks and spalling (U-Net, DeepLabV3+, SegFormer or YOLO segmentation; choose by validation results).
- **Registration:** classical feature matching and homography first. Evaluate learned methods only if classical methods fail on test data.
- **Training data:** combine public structural damage datasets with our own labelled photographs. Target 500 to 1,500 carefully annotated local images for fine-tuning. Annotate with segmentation masks.
- **Augmentation:** lighting, blur, perspective, colour and noise changes to approximate different phones and conditions.
- **Explainability:** every flagged change must be shown visually on the image, not just as a number.
- **Human in the loop:** engineer feedback is logged and used to measure false alarms and improve models.

## 9. Non-functional requirements

- **Accuracy over automation:** prefer sending an uncertain case to human review over silently scoring it.
- **Performance:** analysis of one image pair in under 60 seconds on a single GPU or a reasonable CPU setup.
- **Usability:** a trained capturer can complete a capture point in under two minutes.
- **Offline tolerance:** capture must work without connectivity.
- **Privacy:** blur faces and number plates, store only exterior imagery of buildings, restrict access by role.
- **Auditability:** keep an immutable history of images, model versions, scores and engineer decisions.
- **Cost:** no specialised hardware. Run on commodity phones and open-source tools.

## 10. Success metrics

The numbers below are **targets to be measured, not results**. Do not quote them as achievements.

| Metric | Target | How measured |
|---|---|---|
| Crack segmentation IoU | 0.75 or higher | Against manual masks on a held-out set |
| Spalling segmentation F1 | 0.80 or higher | Same |
| Registration success rate | 90 percent or higher of capture pairs | Fraction of pairs passing alignment threshold |
| Change detection recall on staged worsening | 85 percent or higher | Staged and archival test set |
| False alarm rate on unchanged pairs | 10 percent or lower | Same |
| Ranking quality | Deliberately worsened cases in top 20 percent of queue | 100-image-pair ranking test |
| Engineer agreement | 70 percent or more of flagged items rated useful | Engineer review of a sample |

## 11. Validation plan

Real deterioration is slow, so validation must not depend on waiting for damage.

1. **Controlled surfaces:** panels or walls where marks and damage are progressively added or simulated safely, photographed on a schedule under varied lighting and cameras.
2. **Archival pairs:** find before and after photographs of the same building element from different dates (renovation records, street imagery, society archives) with known change.
3. **Accessible buildings:** with permission, photograph exterior regions of five to ten older buildings repeatedly during the challenge. Do not describe any building as unsafe.
4. **Manual annotation:** one or two people independently annotate a test set, and agreement between annotators is reported.
5. **Ranking test:** mix unchanged and deliberately changed pairs and measure whether changed cases rise to the top.
6. **Engineer review:** ask at least one practising structural or civil engineer to review the outputs and give written feedback.

# 12. Timeline against SFIC 2026

| SFIC stage | Dates | NIRIKSHAN deliverables |
|---|---|---|
| Ideate | to 11 Oct 2026 | Idea submission, problem statement, PRD, validation plan |
| Build and Evaluate | 12 Oct to 10 Nov | Capture app prototype, crack and spalling models, alignment, change metric, priority queue, first staged tests |
| Validate and Test | 11 Nov to 19 Nov | Field captures on accessible buildings, ranking test, engineer feedback |
| Final Select | 20 Nov to 26 Nov | Final report, live demo, results with honest error analysis |

Suggested internal sprints during Build:

- **Week 1:** data collection, annotation setup, baseline crack model, capture overlay prototype
- **Week 2:** spalling model, image registration, first before/after change output
- **Week 3:** priority score, dashboard, building timeline
- **Week 4:** staged tests, error analysis, demo preparation

# 13. Risks and mitigations

| Risk | Impact | Mitigation |
|---|---|---|
| Insufficient labelled spalling and damage data | Weak models | Use public datasets plus local labelling. Narrow MVP to cracks and spalling. |
| Lighting and camera differences cause false change | False alarms | Illumination normalisation, alignment confidence gate, controlled tests across phones |
| Poor capture consistency by users | Unusable comparisons | Overlay guidance, quality gates, recapture prompts |
| Visible distress does not reflect hidden problems | Misleading comfort | State the limitation prominently. Never output a safety verdict. Human review of every ranking. |

| Risk | Impact | Mitigation |
|---|---|---|
| False "low priority" on a deteriorating building | Safety and reputational risk | Conservative thresholds, uncertainty routed to human review, regular engineer audits |
| Building access and ownership disputes | Cannot capture | Use exterior, legally accessible views. Work through MHADA, societies or consenting owners. |
| Not enough time to show real deterioration | Weak validation story | Staged and archival data, clearly labelled as such |
| Scope creep | Unfinished prototype | Enforce P0 only until the core loop works end to end |

## 14. Ethics, privacy and safety

- The product supports engineers and never replaces them.
- Outputs use "inspection priority" language, not safety language.
- Residents are not penalised or publicly named based on automated output.
- Personal data in images is blurred on device. Interior images are out of scope.
- Building scores are visible only to authorised users, to avoid panic or property-value harm from unverified automated outputs.

## 15. Open questions

1. Will MBRRB or any civic body agree to review sample outputs or provide anonymised historical inspection data?
2. Is there a practising structural engineer available to advise and review?
3. Which accessible buildings can the team photograph repeatedly, and with whose permission?
4. What is the actual judging rubric for SFIC, and does it weight working prototypes over concepts?
5. What compute is available for training (local GPU, Colab, institutional cluster)?
6. Which SFIC category and problem statement code should this be filed under? This must be confirmed on the registration portal.

# 16. Appendix: Priority score definition (draft)

P = w1·C + w2·S + w3·R + w4·W + w5·T, scaled to 0 to 100.

- C: normalised change in crack extent between aligned captures
- S: normalised change in spalling area
- R: exposed reinforcement indicator (Phase 2)
- W: moisture deterioration change (Phase 2)
- T: rate of change across recent captures

Initial weights are set by hand with engineer input, then adjusted using feedback. Missing terms are excluded and weights renormalised, so the MVP score uses C, S and T only. Every score displays its contributing factors and the images behind them.